WORKED SOLUTION

STEP II 2005, Question 12

Probability and Statistics • Bayes' Theorem

STEP II 2005, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s2-q12

The key idea

This is Bayes' theorem with a fair coin as the prior. Because the coin is fair, the probability of heads given the answers is $\frac{1}{2}$ exactly when the answers are *equally likely* whether the coin showed heads or tails. So in each part we just compare two likelihoods, $\mathrm{P}(\text{answers} \mid H)$ and $\mathrm{P}(\text{answers} \mid T)$, conditioning on which twin is online to work them out.

Solution

Let H and T be the events that the coin shows heads and tails, so $\mathrm{P}(H) = \mathrm{P}(T) = \frac{1}{2}$. Let A be the event that Anna is online and B the event that Bella is online, with $\mathrm{P}(A) = p$, $\mathrm{P}(B) = q = 1 - p$. Who is online has nothing to do with the coin, so A and B are independent of H. In parts (ii) and (iii) we take the coin to be tossed once and both e-mails to be answered by the same twin, the one online when we write. (See the remark at the end for the other reading.)

Write $t = ap + bq$, the overall probability that a single answer is truthful.

For any record of answers E, Bayes' theorem with $\mathrm{P}(H) = \mathrm{P}(T)$ gives

$$\mathrm{P}(H \mid E) = \frac{\frac{1}{2}\mathrm{P}(E \mid H)}{\frac{1}{2}\mathrm{P}(E \mid H) + \frac{1}{2}\mathrm{P}(E \mid T)} = \frac{\mathrm{P}(E \mid H)}{\mathrm{P}(E \mid H) + \mathrm{P}(E \mid T)}.$$

This equals $\frac{1}{2}$ if and only if $\mathrm{P}(E \mid H) = \mathrm{P}(E \mid T)$. That one observation does most of the work.

Part (i)

Let Y be the event that the answer is 'yes'. If the coin shows heads, 'yes' is the truth. Conditioning on who is online,

$$\mathrm{P}(Y \mid H) = pa + qb = t.$$

If the coin shows tails, 'yes' is a lie, so

$$\mathrm{P}(Y \mid T) = p(1-a) + q(1-b) = (p+q) - (ap+bq) = 1 - t.$$

Hence

$$\mathrm{P}(H \mid Y) = \frac{t}{t + (1-t)} = t = ap + bq.$$

So $\mathrm{P}(H \mid Y) = \frac{1}{2}$ if and only if $ap + bq = \frac{1}{2}$, that is, if and only if $2(ap+bq) = 1$, as required.

Part (ii)

Let E be the event 'yes' then 'no'. Suppose Anna is online. Her two answers are independent, and exactly one of them is true whichever way the coin landed:

$$\mathrm{P}(E \mid H \cap A) = a(1-a) \qquad \text{(truthful 'yes', then a lie)},$$
$$\mathrm{P}(E \mid T \cap A) = (1-a)a \qquad \text{(a lie, then truthful 'no')}.$$

These are equal. In the same way $\mathrm{P}(E \mid H \cap B) = b(1-b) = \mathrm{P}(E \mid T \cap B)$. Therefore

$$\mathrm{P}(E \mid H) = pa(1-a) + qb(1-b) = \mathrm{P}(E \mid T),$$

and by the observation above $\mathrm{P}(H \mid E) = \frac{1}{2}$, whatever the values of p, a and b.

Ask yourself Why do we get exactly $\frac{1}{2}$ here with no condition at all?

Contradictory answers mean that, whatever the coin showed, the twin told the truth once and lied once. That combination is equally likely under heads and under tails, so it carries no information about the coin, and we are left with the prior, $\frac{1}{2}$.

Part (iii)

Now let E be the event 'yes' then 'yes'. Given heads, both answers are true; given tails, both are lies. Conditioning on who is online,

$$\mathrm{P}(E \mid H) = pa^2 + qb^2,$$
$$\mathrm{P}(E \mid T) = p(1-a)^2 + q(1-b)^2.$$

Subtracting, and using $(1-a)^2 = 1 - 2a + a^2$,

$$\begin{aligned}\mathrm{P}(E \mid H) - \mathrm{P}(E \mid T) &= p(a^2 - (1-a)^2) + q(b^2 - (1-b)^2) \\ &= p(2a-1) + q(2b-1) \\ &= 2(ap+bq) - (p+q) \\ &= 2(ap+bq) - 1.\end{aligned}$$

So if $2(ap+bq) = 1$, the two likelihoods are equal and $\mathrm{P}(H \mid E) = \frac{1}{2}$, as required.

(In fact the calculation shows more: the difference of the likelihoods is exactly $2(ap+bq) - 1$, so $\mathrm{P}(H \mid E) = \frac{1}{2}$ holds *if and only if* $2(ap+bq) = 1$, the same condition as in part (i).)

Remark: the other reading

If instead whichever twin happens to be online is chosen afresh for each e-mail, independently, then each answer is truthful with probability t, independently. Then in (ii) $\mathrm{P}(E \mid H) = t(1-t) = \mathrm{P}(E \mid T)$, and in (iii) $\mathrm{P}(E \mid H) = t^2$ and $\mathrm{P}(E \mid T) = (1-t)^2$, which are equal when $t = \frac{1}{2}$. The conclusions are the same.

Checks

- If both twins are always truthful ($a = b = 1$), then $t = 1 \neq \frac{1}{2}$, and in (i) a 'yes' makes heads certain: $\mathrm{P}(H \mid Y) = t = 1$. Contradictory answers as in (ii) are then impossible, which is consistent with $a(1-a) = b(1-b) = 0$.
- With $p = 0.3$, $a = 0.8$ and $b = \frac{0.5-0.24}{0.7} \approx 0.371$ (so $2(ap + bq) = 1$), a simulation of 2×10^5 rounds in the same-twin model gave $\mathrm{P}(H \mid \text{yes, yes}) \approx 0.496$. Exact enumeration with fractions over thousands of random (p, a, b) confirmed all three results in both readings.
- If $p = \frac{1}{2}$ and $a = b = 0.9$, the condition fails and two 'yes' answers give $\mathrm{P}(H \mid E) = \frac{81}{82}$, far from $\frac{1}{2}$, so the condition in (iii) really matters.

Common slips

- Writing $\mathrm{P}(Y \mid T) = 1 - \mathrm{P}(Y \mid H)$ without justification. It happens to be true here, but only because a 'yes' under tails is a lie; say so.
- In (iii), treating the two answers as independent truthful answers with probability t each while also assuming the same twin answers both. Given the twin they are independent, but unconditionally they are not, since both depend on who is online.

Beyond STEP

The single principle here is that the posterior odds equal the prior odds times the likelihood ratio: evidence that is equally likely under both hypotheses tells you nothing. The same-twin model is a simple *mixture*: answers are independent given a hidden variable (who is online) but dependent overall, an idea at the heart of hidden-variable and Bayesian models. Blitzstein and Hwang, *Introduction to Probability*, has an excellent treatment of conditioning and Bayes' rule in exactly this spirit.

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